VENUS ATMOSPHERIC FLASH INVESTIGATOR

VAFI

Observer quick manual | Detector version 7 | October 28, 2026

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Introduction

When our planet passes through a meteoroid stream, we observe a meteor shower from the ground or from space, as meteoroids′ high speeds interact with the atmosphere. The same phenomenon can occur on other terrestrial planets with atmospheres, such as Mars and Venus, and several candidate minor bodies have been suggested as the parent bodies of meteor streams on these planets, including Halley-type comets, long-period comets, and known meteoroid streams of Encke or Jupiter-family type. Indeed, the speeds of meteoroids in Venus′s atmosphere are sufficient to generate meteors. At the heliocentric distance of Venus of about 0.7 au, the escape speed of a body belonging to the Solar System is about 50.4 km/s, and the planet itself has an orbital speed of 33.9 km/s and an escape speed of about 10.4 km/s. With these numbers, the planetocentric speed of interplanetary meteoroids belonging to the Solar System is between 10.4 and 85 km/s, a range not significantly different from the Earth′s meteor range of 11.4 and 72 km/s. Given the brightness of Venus's disk, one can only hope to detect fireballs in its atmosphere by observing the planet's night hemisphere. To search for potential fireballs,

software is required to analyze images and videos captured with a telescope, typically at an image scale of around 0.1 arcsec/pixel.

Purpose

VAFI is a Python program for searching for brief luminous events on the nightside of Venus in telescope observations. It reads BMP or FITS image sequences and AVI video recordings. VAFI produces candidates for verification; a detection is not, by itself, proof of a Venusian fireball.

How the detection works

01 Venus geometry: The positions of Earth and Venus are computed from fixed Keplerian orbital elements at the observation date (UTC). The Sun-Venus-Earth phase angle, illuminated fraction, and apparent diameter are calculated. Using the image scale (arcsec/pixel), an analytical model of the illuminated Venus disk and its terminator is generated.

02 Mask creation: In manual mode, the theoretical Venus profile is superimposed on the first image in an interactive window. The user can translate, rotate, and resize the profile to obtain the best fit to the observed disk. The accepted geometry is saved in a JSON file and can be reused in subsequent analyses. Automatic intensity-based, geometric, and phase-based masking modes are also available. The bright region, terminator margin, limb margin, and glare-contaminated pixels are excluded. Only the remaining green nightside area is searched.

03 Frame registration: Since telescope tracking errors cause Venus to shift between consecutive frames, all images are automatically registered to the first frame using template matching on the bright hemisphere. Subpixel translations are applied, and poorly aligned frames are excluded from the analysis.

04 Flash search: The Venus-fitted theoretical profile is used to exclude the illuminated hemisphere, a region around the terminator, and the planetary limb. Additional masking reduces contamination from scattered light. Only the remaining night side pixels are searched for flashes. The algorithm searches for local brightness maxima within 3×3-pixel regions that exceed the brightness of the eight neighboring pixels by at least N standard deviations. Each candidate must also exceed the temporal background at the same registered position by at least M standard deviations. The analysis is repeated using spatial averaging over different pixel blocks (1x1, 2×2, 3×3, ...) and temporal averaging over different numbers of consecutive frames, allowing the detection of flashes with different angular sizes and durations. Version 7 also tests for a connected, positive multipixel residual.

05 Candidate review: Detections close in position and time are merged into individual candidate events. The program produces CSV and text tables with frame numbers, timestamps, coordinates, detection significance, and signal excess, together with diagnostic images, alignment information, and the adopted Venus mask.

Before the first run

- Use Python 3.10 or later on Windows. Install NumPy, OpenCV and Astropy (Astropy is needed for FITS input).

```
py -m pip install numpy opencv-python astropy
```

- Place VAFI_v7.py in your working folder, or supply its full path. VAFI is the software name; the example commands use the current v7 script filename.
- Record the true first-frame UTC time, frame interval or video FPS, and the actual image scale (arcsec/pixel). Do not copy the example scale unless it applies to your data.

RECOMMENDED WORKFLOW: Always create and inspect a manual-mask preview first; then reuse the accepted JSON geometry to analyze the full sequence. Save each analysis in a separate results folder.

Quick start: create the manual mask, then search

The commands below are examples for the Windows command line. Run each as one command; replace paths, UTC date/time, and plate scale with your observation values. Example scale 0.088 arcsec/pixel is specific to one observing setup.

Step A — preview and approve the Venus mask

Example for an AVI recording (frame cadence is read from the AVI FPS metadata):

```
py VAFI_v7.py --input ".\Venus.avi" --output ".\MaskPreview" --start-utc 2026-07-05T17:34:00 --plate-scale 0.088 --mask-mode manual --preview-only
```

An interactive Tkinter window opens, see Fig. 1. For the record, Tkinter is the standard graphics library built into Python for creating user interfaces (GUIs). The left view shows the first frame; the right view overlays the analytical illuminated silhouette and the safe nightside. Drag the right-hand overlay to translate it; use the controls to rotate or adjust the radius. The canvas scrollbars preserve the image aspect ratio.

Manual-fit control	Action
Mouse drag on the right view	Move the phase profile over Venus
Center X / Y; Radius; Sun angle	Adjust disk position, size, and orientation
A / D; + / −; I / J / K / L	Rotate; resize; shift by keyboard
S / Space / Enter	Accept and save the fitted geometry
Esc	Cancel the manual fit

After acceptance, inspect MaskPreview\phase_search_mask.png and MaskPreview\disk_and_mask.png. Green is the searchable nightside. If the green area is wrong, repeat the preview before proceeding.

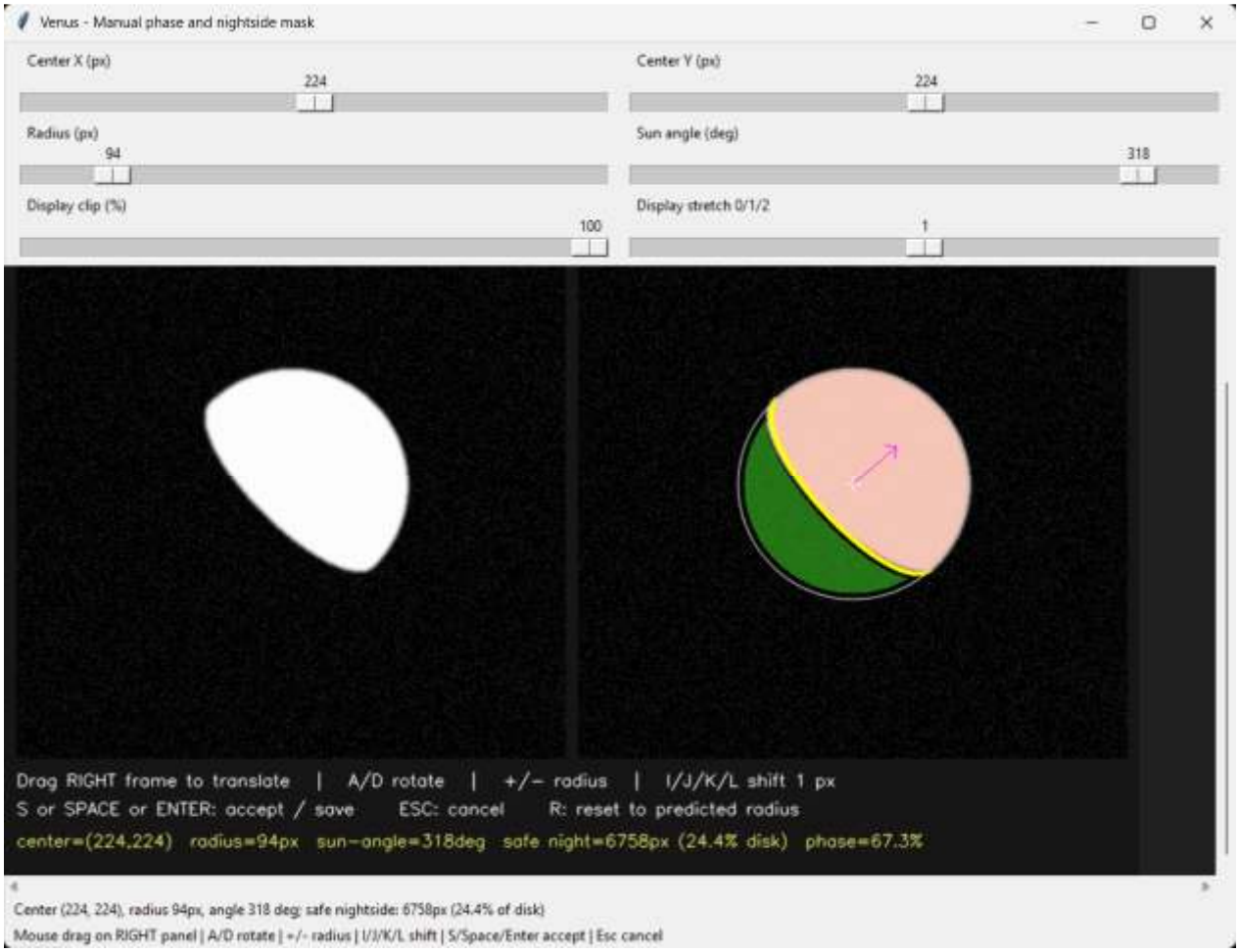


Fig. 1 – The VAFI interactive Tkinter window to create the measure mask for fireball search. The controls are self-explanatory; you can adjust the position of the measurement mask, as well as the radius and the direction of incoming sunlight.

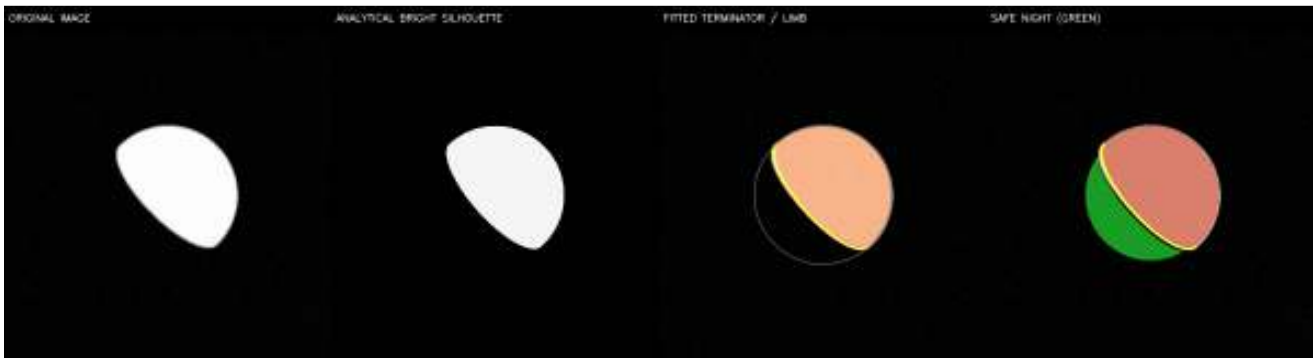


Fig. 2 – From left to right, the original Venus image (in this example from a synthetic BMP sequence), the analytical Venus disk computed from VAFI based on the input date, the computed terminator mask, and the fireball search region in green.

Step B — run the full analysis with the saved mask

Once the measurement mask has been saved, VAFI can be run again to begin the analysis for detecting fireballs in the night-side hemisphere of Venus, utilizing the measurement mask created previously.

```
py VAFI_v7.py --input ".\Venus.avi" --output ".\Results_AVI" --start-utc 2026-07-05T17:34:00 --plate-scale 0.088 --mask-mode manual --manual-config ".\MaskPreview\manual_mask_geometry.json" --spatial-scales 1,2 --time-scales 1,2 --N 1.5 --M 7 --min-psf-pixels 2 --pixel-support-sigma 2.5
```

SAVED-MASK CONSTRAINT: The JSON is tied to the original first image, phase/date, plate scale, and mask margins. Keep --bright-margin, --limb-margin, and --phase-glare-fraction identical to the preview settings; changing any of them requires a new approved fit.

Use a BMP or FITS sequence instead of an AVI

BMP: replace --input with the BMP folder and provide an actual frame interval when --start-utc is used. Example:

```
py VAFI_v7.py --input ".\BMP_frames" --output ".\Mask_BMP" --start-utc 2026-07-06T16:53:15 --cadence 0.5 --plate-scale 0.088 --mask-mode manual --preview-only
```

FITS: replace --input with a folder containing FITS files. If DATE-OBS is present in the first file, VAFI can use it for the phase calculation and can read per-image observing times:

```
py VAFI_v7.py --input ".\FITS_frames" --output ".\Mask_FITS" --plate-scale 0.088 --mask-mode manual --preview-only
```

For BMP/FITS analysis, repeat the input command without --preview-only, choose a new output folder, and add --manual-config pointing to the JSON produced in the matching preview. Do not mix BMP and FITS files in one input folder.

Command-line reference | inputs, time, and masking

Arguments are grouped by their function. Values in the “Default” column are the script defaults; “—” indicates a required value or no preselected value.

Files, timing and preview

Option	Default	Meaning
--input	required	BMP-only or FITS-only folder, or one .avi file.
--output	required	Folder for mask previews, diagnostic files and detection results.
--start-utc	—	Start time of the first frame, UTC ISO format; also supplies phase date unless overridden.
--cadence	auto / —	Seconds between frames. AVI uses FPS automatically; BMP needs cadence if start-utc is supplied.

Option	Default	Meaning
--observation-utc	—	Override UTC date/time for Venus geometry only (not frame timestamps).
--preview-only	off	Build first-frame mask and exit without processing all frames.
--max-cutouts	60	Maximum number of candidate PNGs written to event_cutouts/.

Manual and analytical phase mask

Option	Default	Meaning
--mask-mode	intensity	intensity, geometric, phase or manual. Manual is recommended for difficult Venus imagery.
--plate-scale	—	Arcseconds per pixel; required in phase or manual mode.
--manual-config	—	Load approved manual_mask_geometry.json; skips the fitting window.
--bright-margin	10 px	Extra excluded pixels at illuminated side / terminator. Larger = safer but smaller search area.
--limb-margin	5 px	Exclude an outer rim of the Venus disk.
--phase-glare-fraction	0.65	Veto exceptionally bright pixels in phase/manual masks; 1 disables the extra veto.
--cx, --cy	—	Optional disk center in original pixel coordinates.
--radius	—	Override physical radius in pixels; verify plate scale before using.
--phase-image-angle	—	Sunward image direction in degrees (0=right, 90=down).
--phase-fit-profile	auto	In phase mode: auto, binary or lambert template.
--phase-min-fit-score	0.30	Minimum accepted automatic phase-fit correlation for a full run.

Other mask modes / advanced geometry

Option	Default	Meaning
--disk-threshold-fraction	0.12	Intensity level used for automatic disk estimate (non-phase modes).
--bright-fraction	0.03	Brightness cutoff for intensity/geometric masking.
--night-cap	0.25	Conservative nightside projection threshold in geometric mode.
--crop-pad	24 px	Margin retained around the disk in the registered data cube.

Command-line reference | registration and detection

Image alignment

Option	Default	Meaning
--registration-method	template	Align bright crescent using template matching; phase is an alternative registration method.
--align-size	512 px	Longest side of resized registration frame; original data are stored separately.
--min-registration-response	0.7 / 0.03	Reject uncertain alignment: default 0.7 for template, 0.03 for phase registration.

Spatial and temporal candidate selection

Option	Default	Meaning
--spatial-scales	1,2,3	Square averaging blocks in pixels. Example 1,2 searches native resolution and 2×2 bins.
--time-scales	1,2,4,16,32	Numbers of successive frames averaged; use scales appropriate to expected flash duration.
--N	1.5	Spatial significance versus eight surrounding pixels. Higher rejects more local peaks.
--M	7	Temporal significance versus neighboring reference windows. Higher rejects more temporal noise.
--neighbors	6	Number of comparison windows on each side of test interval.
--min-references	6	Minimum usable temporal reference windows required.
--noise-floor	auto	Lower bound on estimated per-pixel RMS noise (image intensity units).
--min-excess	0	Minimum positive intensity difference; checked for spatial and temporal excess.
--min-psf-pixels	2	Minimum connected positive residual pixels in original-resolution image. 1 disables this test.
--pixel-support-sigma	2.5	Significance required for each pixel counted toward the connected-pixel filter.
--max-spatial-candidates	300	Keep up to this many local peaks per frame and spatial scale.

Event consolidation

Option	Default	Meaning
--merge-frames	3	Maximum frame separation for merging detections into one event.
--merge-radius	8 px	Maximum positional separation for merging detections into one event.

PRACTICAL STARTING POINT: For real observations, begin with N=1.5, M=7, min-psf-pixels=2 and pixel-support-sigma=2.5, then tune against known injected flashes and false triggers. Values optimized for one camera/PSF/cadence should not be adopted blindly for another.

Where to find the results

VAFI writes files in the folder specified by --output. A preview-only run creates mask/geometry diagnostics but no candidate tables. A complete run additionally writes alignment and candidate files.

File or folder	Contents and use
venus_candidates.csv	Machine-readable candidate list: UTC/frame, x/y, detection scales, spatial_sigma, temporal_sigma, signal_excess, support_pixels and more.
venus_candidates.txt	Compact, human-readable candidate list; its Excess column is a signal increase in image units, not σ.
event_cutouts/V####.png	Three-panel event diagnostics: registered frame; temporal difference; actual masked search area.
alignment.csv	Per-frame registration dx/dy, quality, and usable flag; for AVI includes video_time_s.
disk_and_mask.png	Planet location and current effective nightside mask; check green searchable region.
phase_model_overlay.png	An analytical, illuminated phase and a fitted outline of the observation.
phase_search_mask.png	Color display of the geometric mask (phase/manual modes).
phase_bright_silhouette.png	Calculated illuminated silhouette alone (phase/manual modes).

File or folder	Contents and use
manual_mask_geometry.json	Reproducible approved manual fit to reuse using --manual-config.
manual_fit_preview.png	Complete image/overlay accepted in the manual fitting GUI.
phase_geometry.txt	Keplerian phase, apparent diameter, and fitted geometry details.
mask_diagnostics.txt	Pixel counts and mask checks, including usable nightside area.

Interpreting the three-panel candidate cutout

In the output folder, there is the “event_cutouts” folder containing PNG images of all candidate fireball detections for a fast visual inspection. Each PNG image is divided into three parts, see Fig. 3. Left: the registered image at the candidate peak, with a white cross. Center: the frame minus a median background from neighboring frames, displayed with a relative false-color scale (not a σ significance map). Right: the actual mask used to search: green = searchable nightside; red = excluded illuminated/glare region; orange = excluded portion inside the planetary disk; gray = outside the disk. The cross should fall on green. The candidate fireballs list is in the output folder in the file **venus_candidates.txt,** that show the images number and the pixel position of each detection.

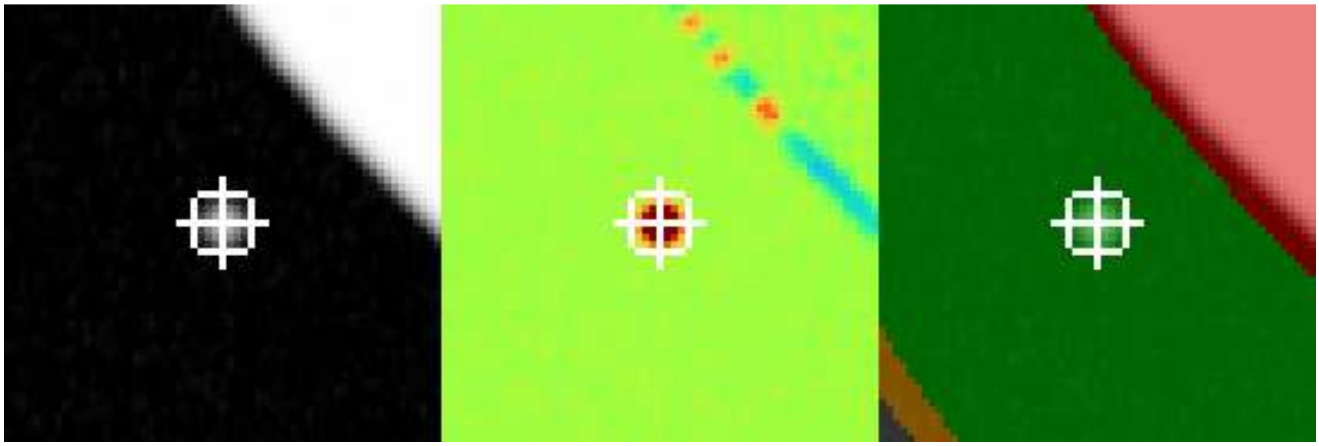


Fig. 3 – Detection of a synthetic fireball in a BMP image sequence.

Observer checks before reporting a candidate

- Verify that the saved mask follows Venus and that the candidate lies in green, away from unsafe bright boundaries.
- Check alignment.csv for rejected/poorly registered frames near the event. Review the original frames for hot pixels, cosmic rays, seeing, and video compression artifacts.
- Record the original data source, UTC start, cadence/FPS, plate scale, mask margins and all detection thresholds. Prefer independent recordings from a second observatory.

SCIENTIFIC INTERPRETATION: A missing event very near the terminator or limb may lie outside VAFI’s searchable area. Quote detection completeness only for injected events actually inside the effective green region, and evaluate false positives separately. VAFI candidates require human review and ideally independent confirmation.